

Exhibit 101

U.S. Patent No. 10,217,523 Domestic Industry Claim Chart (Samsung)

EXHIBIT 101

Domestic Industry Claim Chart for Claims of U.S. Patent No. 10,217,523 (the “’523 Patent”) Based on Samsung Practicing ’523 Products

As shown in the chart below, the Samsung Practicing ’523 Products, which include Samsung DDR5 MRDIMMs, practice at least claim 1 of the ’523 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Samsung Practicing ’523 Products do not practice the ’523 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Samsung Practicing ’523 Products, such differences are insubstantial and the Samsung Practicing ’523 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Samsung Practicing ’523 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Samsung Practicing ’523 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

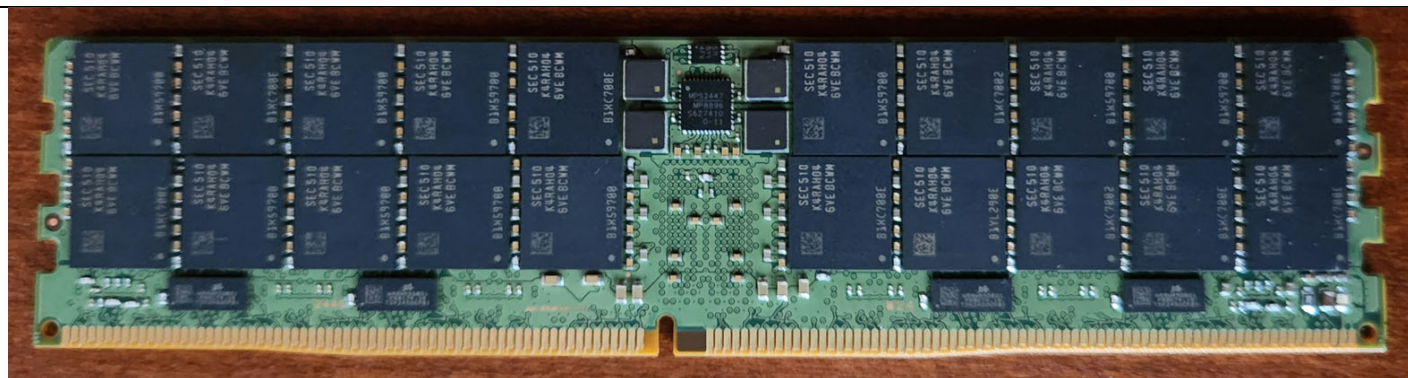
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Claim Limitation	Evidence of Use
[1.a] A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:	Samsung's DDR5 MRDIMM meets all of the limitations of claim 1. To the extent that the preamble is limiting, Samsung's DDR5 MRDIMM is "[a] memory module accessible in a computer system by a system memory controller via a system memory bus." Below are images of the front and back of a Samsung DDR5 MRDIMM, which is a Multiplexed Rank Dual In-line Memory Module.  Ex. CC-111 (Samsung MRDIMM M327R8GA0EB0-CLVXB).

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Ex. CC-112 (20250811_151215.jpg).



Ex. CC-113 (20250811_151233.jpg).

First, Samsung's MRDIMMs (also called MCRDIMMs¹) are each a dual inline *memory module*.

¹ See, e.g., Ex. CC-62 (Montage HK Proof) at 43 (emphasis added) (describing “MRDIMM” as “multiplexed rank dual in-line memory module, a high-bandwidth server memory module based on the DDR5 LRDIMM architecture adopting the ‘1+10’ design consisting of one MRCD chip and 10 DB chips, which, compared with RDIMM and LRDIMM, enables simultaneous access to two memory arrays on the module and effectively doubles the bandwidth; *the first generation of MRDIMM is also referred to MCRDIMM by some manufacturers.*”).

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For example, MCRDIMM stands for “Multi-Ranked Buffered Dual In-Line Memory Module.” *See* Ex. CC-114 (Samsung MCRDIMM Tech Blog). MRDIMM stands for “Multiplexed Rank Dual Inline Memory Module.” *See, e.g.*, Ex. CC-50 (TrendForce News MRDIMM-MCRDIMM); Ex. CC-62 (Montage HK Proof) at 43 (“MRDIMM” is defined as “multiplexed rank dual in-line memory module . . .”).

As another example, Samsung’s DDR5 MRDIMM uses components made by Montage Technology and Renesas, including Montage’s DDR5 Multiplexed Rank Registering Clock Driver (MRCD) and Multiplexed Rank Data Buffer (MDB) chips.



Ex. CC-113 (20250811_151233.jpg).

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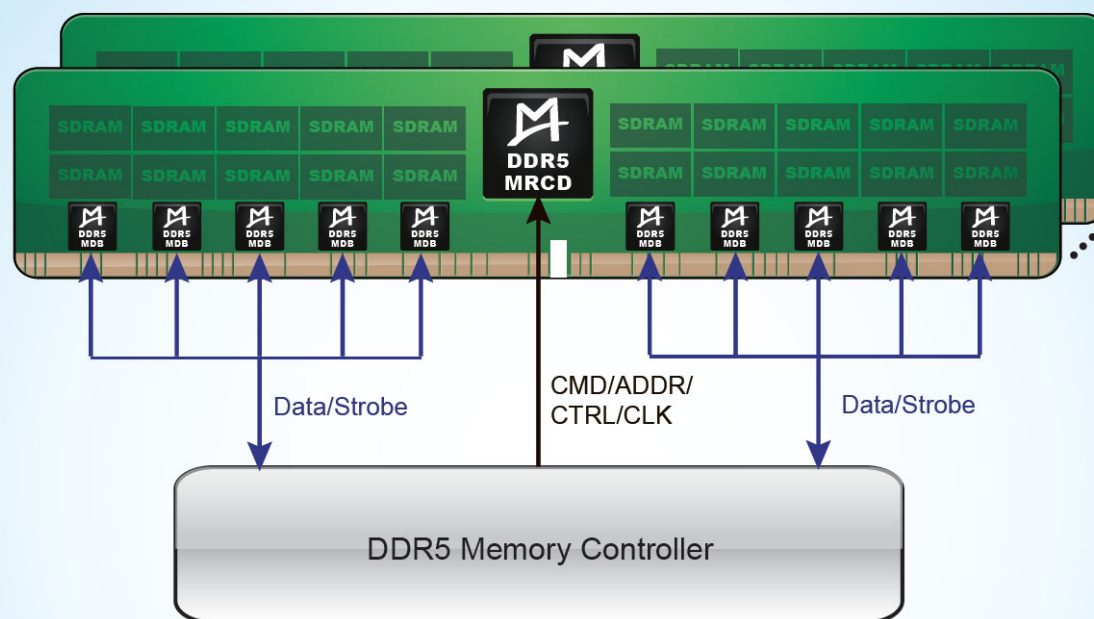
Ex. CC-115 (20250811_151613.jpg).

Montage states that its “MRCD chip is applied in next-generation DDR5 MRDIMM/MCRDIMM *memory modules*” See Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) (emphasis added).

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General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in high-performance computing scenarios, such as AI training, inference, and large-scale data analysis.



[REDACTED]

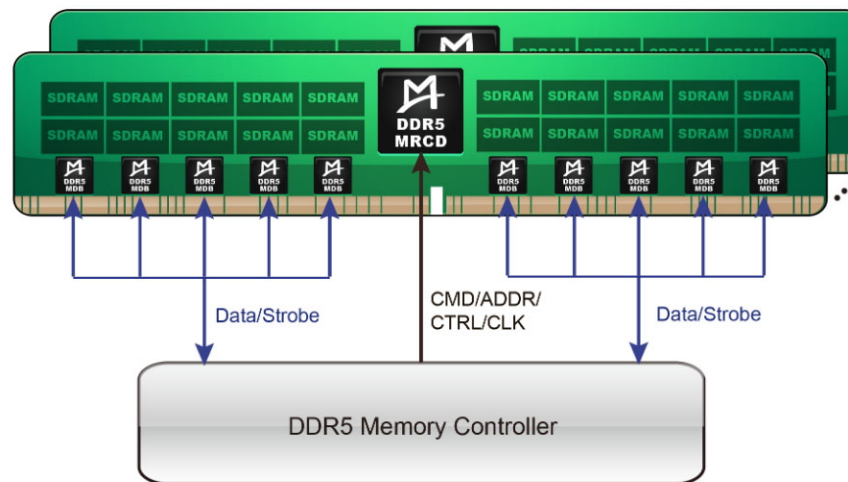
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	Second, the MRDIMM memory module is accessible in a computer system by a system memory controller via a system memory bus (e.g., as a pathway between a CPU and DRAMs for data communication, which would go through a memory slot in the case of communication between a memory controller and a DIMM). According to Ex. CC-114 (Samsung MCRDIMM Tech Blog) (emphasis added): Samsung has developed a Multi-Ranked Buffered Dual In-Line Memory Module (MCRDIMM), which provides more memory and bandwidth <i>without increasing memory slots on the server board</i>. <i>MCRDIMM enables CPU to access two ranks of memory simultaneously on a single DIMM</i> with a chip-set layout of two identical DDR5 DIMMS. The benefit of this approach is that the DRAM devices are not expected to operate at a faster clock frequency. By accessing two DRAM devices at the same time, the CPU can effectively double the memory bandwidth coming from the DIMM. Accessing the two DRAM devices is accomplished by using a special mux data buffer resident on the DRAM, which allows for simultaneous access to both DIMMs thereby <i>doubling the data rate to the host</i>. <i>Samsung's first-generation MCRDIMMs offer data transfer rates of up to 8.8 Gb/s</i>. Samsung is currently sampling 16Gb Mono MCRDIMM devices with enhanced performance, capacity, and power consumption. With this innovative memory solution, Samsung is advancing the next frontier for Artificial Intelligence (AI), Machine Learning (ML), and Large Language Model (LLM) processing. This shows that MRDIMMs/MCRDIMMs are designed to be accessible in a computer system by a system memory controller in a CPU. As another example, Montage describes its MRCD chip as performing the function of “buffer[ing] and re-driv[ing] the address, command, clock, and control signals <i>from the memory controller</i>” to “support[] data rates of up to 8800MT/s.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer).
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General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in high-performance computing scenarios, such as AI training, inference, and large-scale data analysis.



(showing data being exchanged between MRDIMM and DDR5 memory controller).

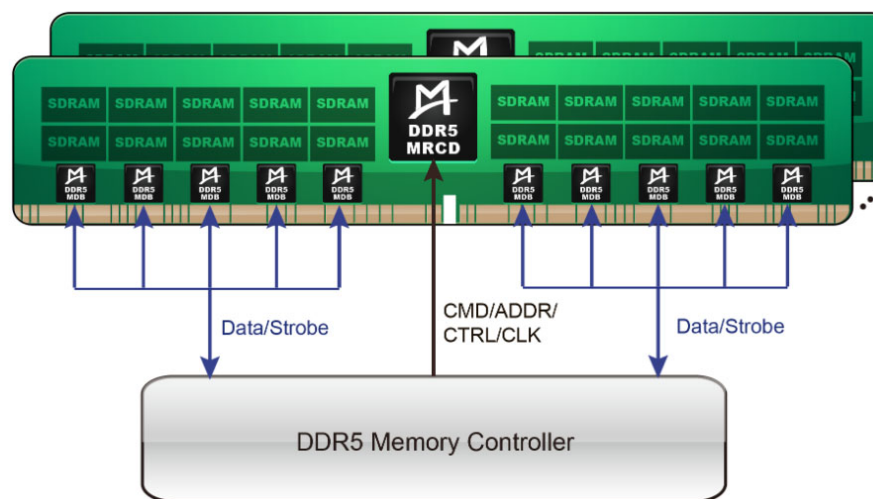
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Likewise, the web page for Montage's MDB states the "MDB chip buffers and re-drives data signals from the memory controller or DRAM devices" and it "efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal." Ex. CC-22 (Montage M88MR5DB01):

Product Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.



Montage's press releases similarly state that its "MRCD chip handles buffering and re-driving of address, command, clock, and control signals, while the ***MDB chips manage data traffic between the memory controller and DRAM chips.***" See Ex. CC-66 (Montage Press Release) (emphasis added):

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The innovative MRCD and MDB chips are fundamental to MRDIMM operation, featuring a “1+10” buffer architecture where each MRDIMM module incorporates one MRCD with ten MDB chips. In this configuration, the MRCD chip handles buffering and re-driving of address, command, clock, and control signals, while the MDB chips manage data traffic between the memory controller and DRAM chips. This architecture, coupled with double data rate and time-division multiplexing technology, enables simultaneous operation of two memory ranks at standard rates, effectively doubling data throughput.



These descriptions by Montage, the DDR5 interface chip supplier for Samsung’s MRDIMMs, again demonstrate that an MRDIMM, like that of Samsung’s MRDIMM that incorporates Montage’s MRCD and MDB chips, is designed to be accessible in a computer system by a system memory controller via a system memory bus.

[1.b] [i] memory devices mounted on a circuit board,
[ii] the memory

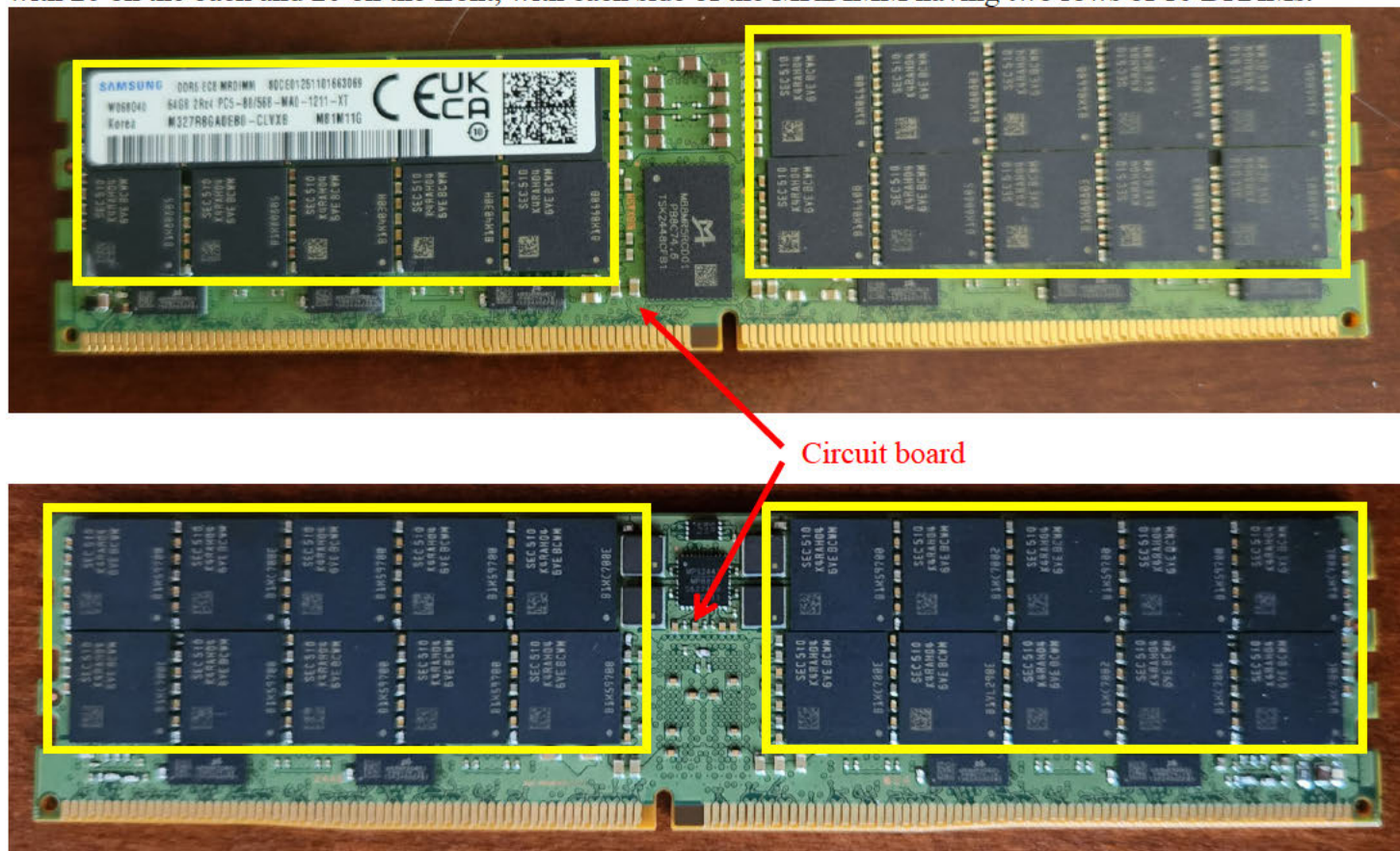
[i] Each Samsung MRDIMM includes memory devices mounted on a circuit board (also called a module board or a printed circuit board or PCB).

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devices having address and control ports and data ports;

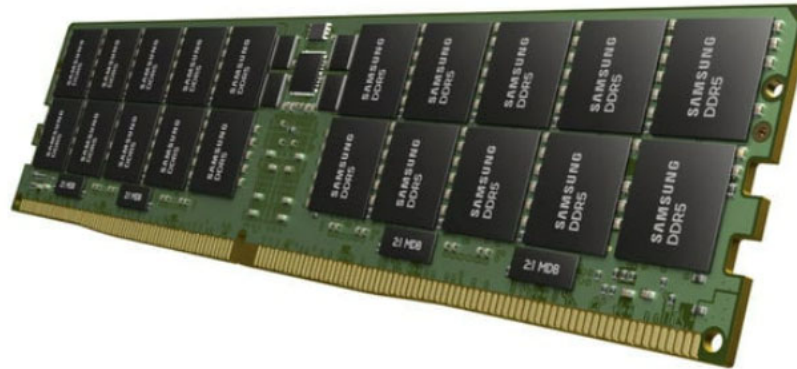
As shown below, there are 40 total DRAMs (highlighted in yellow boxes) mounted on a circuit board (the green PCB) with 20 on the back and 20 on the front, with each side of the MRDIMM having two rows of 10 DRAMs.



Ex. CC-111 (Samsung MRDIMM M327R8GA0EB0-CLVXB) (annotated); Ex. CC-112 (20250811_151215.jpg) (annotated).

See Ex. CC-114 (Samsung MCRDIMM Tech Blog):

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Samsung has developed a Multi-Ranked Buffered Dual In-Line Memory Module (MRDIMM), which provides more memory and bandwidth without increasing memory slots on the server board. This module doubles the bandwidth of existing DRAM components by combining two **DDR5** components to deliver data transmission speeds of up to 8.8 Gb/s.

MRDIMM Features and Benefits

MRDIMM enables CPU to access two ranks of memory simultaneously on a single DIMM with a chip-set layout of two identical DDR5 DIMMS. The benefit of this approach is that the DRAM devices are not expected to operate at a faster clock frequency. By accessing two DRAM devices at the same time, the CPU can effectively double the memory bandwidth coming from the DIMM. Accessing the two DRAM devices is accomplished by using a special mux data buffer resident on the DRAM, which allows for simultaneous access to both DIMMs thereby doubling the data rate to the host.

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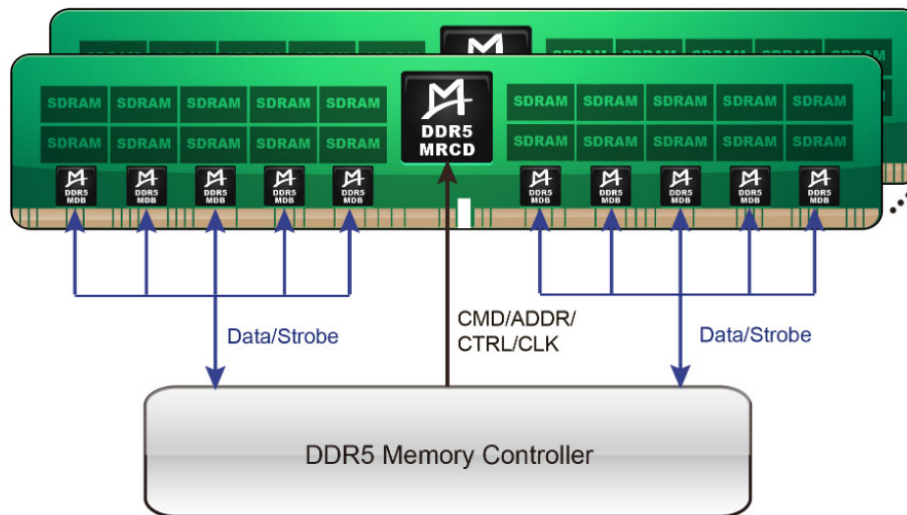
[ii] The memory devices have address and control ports and data ports. For example, the MRCD01 chip from Montage² is designed to “buffer[] and re-drive[] the address, command, clock and control signals from the memory controller.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer). These signals are understood to be driven to at least the memory devices in order to perform “efficient memory management operations.” This implies that the memory devices have address and control ports, which are used to receive the address/control signals re-driven from the MRCD.

General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in high-performance computing scenarios, such as AI training, inference, and large-scale data analysis.

² The MRCD01 chip is an exemplary memory device. Other generations (e.g., 02) would be expected to have the same (or substantially the same) functionality and architectural configuration in the relevant part.

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(showing command/address/control/clock signals being driven to MRCD)

- Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system
- Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs

(“sub-channels” and “pseudo-channels” imply memory devices, and the separately generated CA outputs are sent to memory devices)

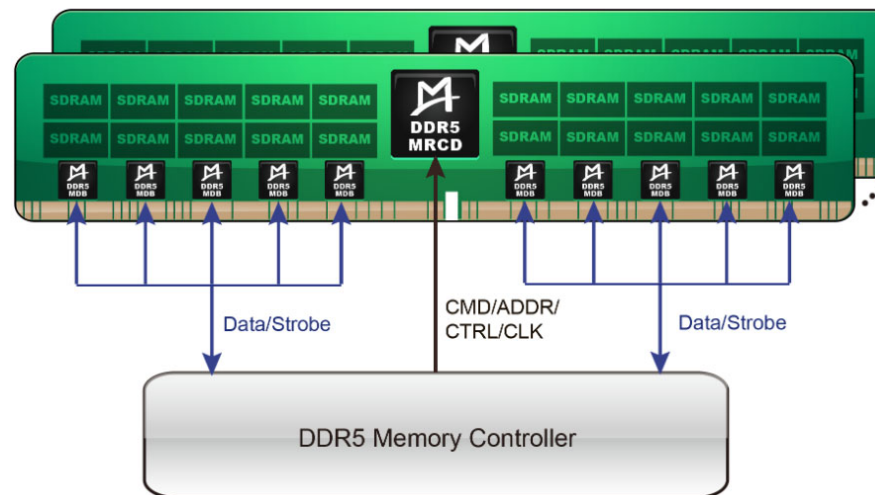
As another example, Montage’s MDB states the “MDB chip buffers and re-drives data signals from the memory controller or DRAM devices” and it “efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01). This implies that memory devices or DDR5 SDRAMs have data ports for receiving and outputting data signals.

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Product Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.

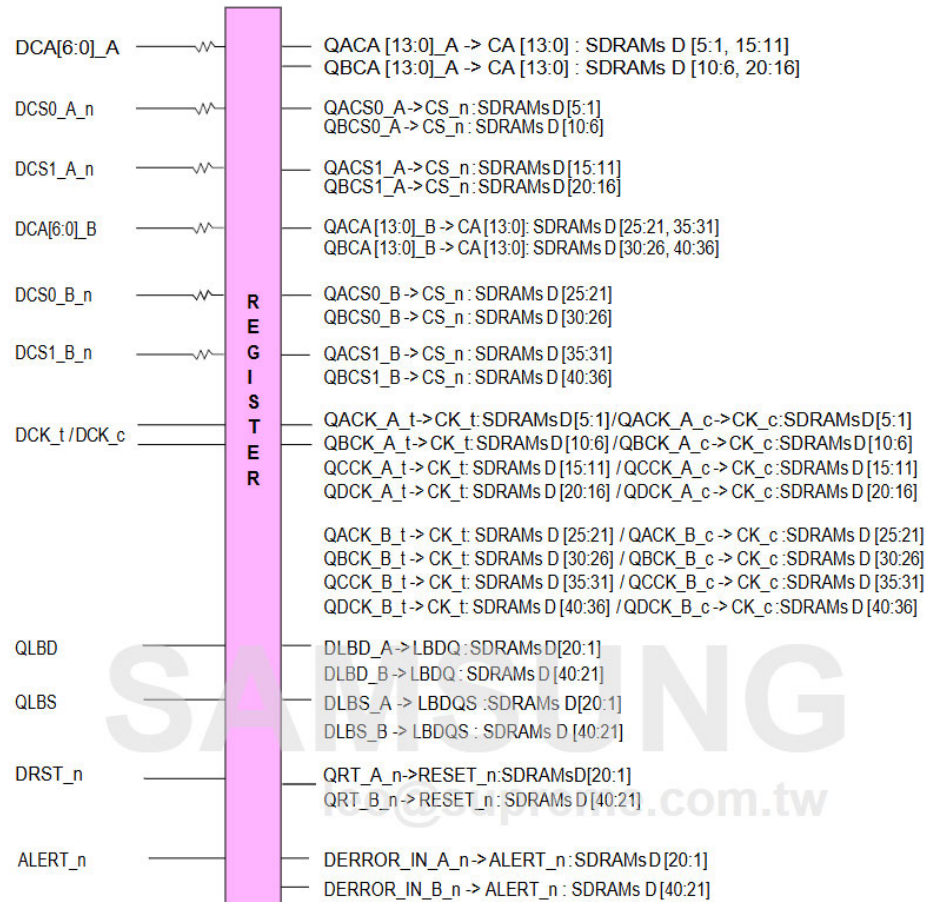


Hence, the memory devices have control/address ports and data ports.

Additionally, it is expected that the same type of SDRAMs would be used on MRDIMMs and RDIMMs. Ex. CC-13 (JEDEC DDR5 MRDIMM Plan) (stating that one planned feature for MRDIMMs is “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption.”). As

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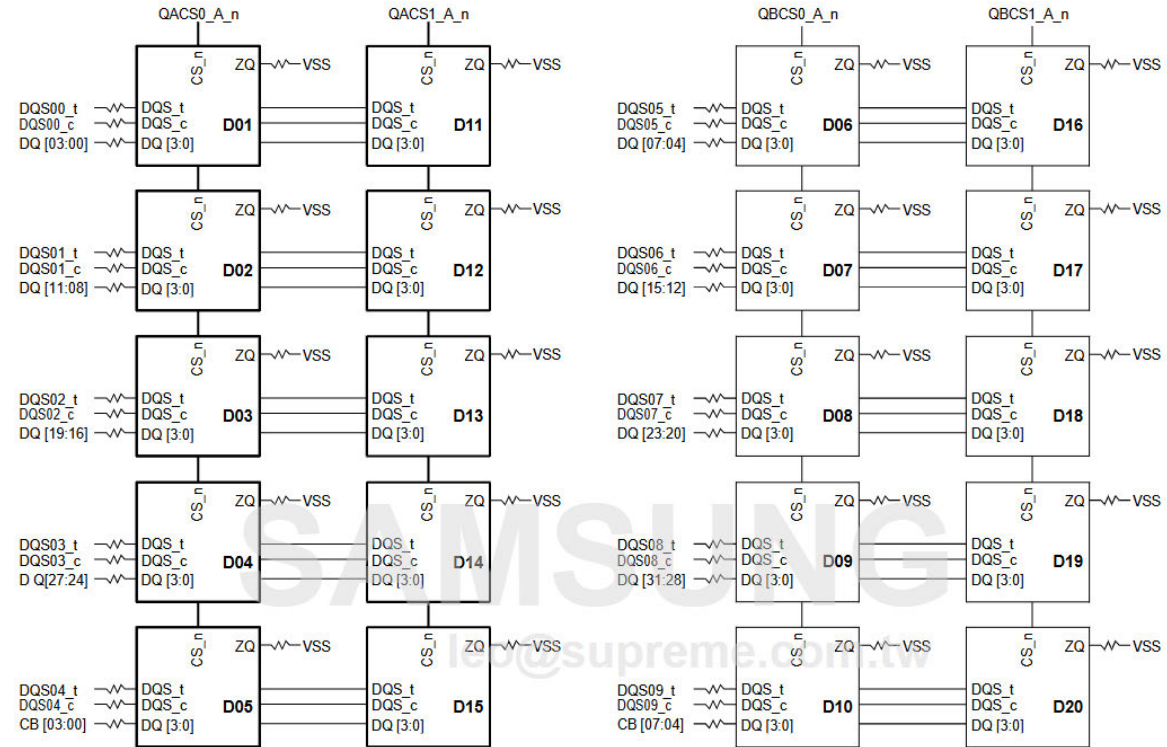
Samsung's DDR5 RDIMM Datasheets illustrate, each of the SDRAM units includes inputs for data signals and address/control signals. *See, e.g.,* Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33–35:



Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33.

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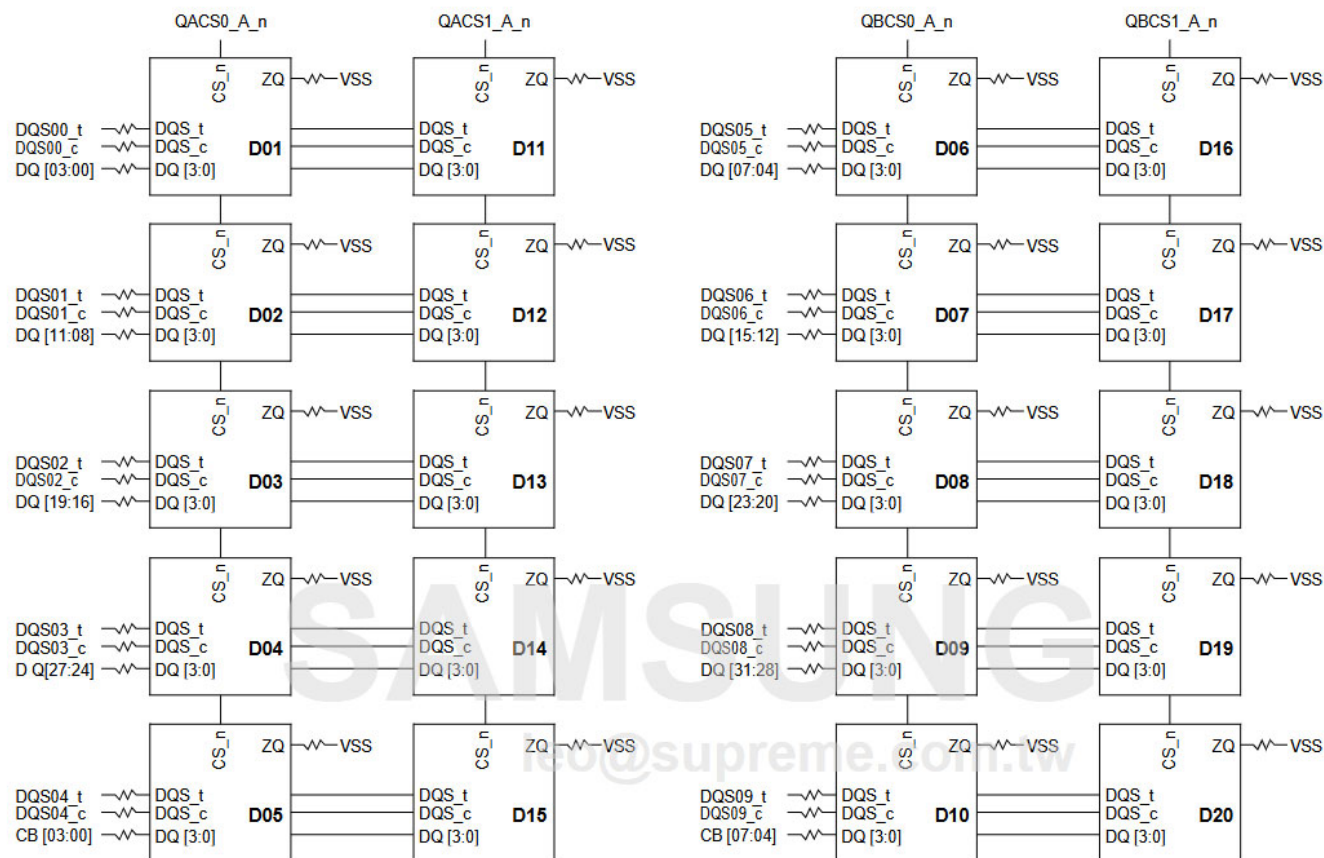
Channel A



Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 34.

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Channel B



Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 35.

See also Ex. CC-5 (JESD79-5C.01_v1.31) at Section 2.4: The definition of the signals (DQ, DQS, CA, CK) and the corresponding ball/signal assignments show that the memory devices have address and control ports (CA pins), as well as data ports (DQ/DQS pins).

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2.4 DDR5 SDRAM x4/8 Ballout Using MO-210

Table 1 — DDR5 SDRAM x4/8 Ballout Using MO-210

AN	1	2	3	4	5	6	7	8	9	10	11
AL		1	2	3	4	5	6	7	8	9	
A	DNU	LBDQ	VSS	VPP				ZQ ⁷	VSS	LBDQS	DNU
B		VDD	VDDQ	DQ2				DQ3	VDDQ	VDD	
C		VSS	DQ0	DQS _t				DM _n ⁴ , TDQS _t ²	DQ1	VSS	
D		VDDQ	VSS	DQS _c				TDQS _c ³	VSS	VDDQ	
E		VDD	DQ4 ¹	DQ6 ¹				DQ7 ¹	DQ5 ¹	VDD	
F		VSS	VDDQ	VSS				VSS	VDDQ	VSS	
G		CA_ODT	MIR	VDD				CK _t	VDDQ	TEN ⁵ (CS1 _n ⁵)	
H		ALERT _n	VSS	CS _n				CK _c	VSS	VDD	
J		VDDQ	CA4	CA0				CA1	CA5	VDDQ	
K		VDD	CA6	CA2				CA3	CA7	VDD	
L		VDDQ	VSS	CA8				CA9	VSS	VDDQ	
M		CAI, ZQ1 ⁸	CA10	CA12				CA13	CA11	RESET _n	
N	DNU	VDD	VSS	VDD				VPP	VSS	VDD	DNU

NOTE 1 DQ4-DQ7 are higher order DQ pins and are not connected for the x4 configuration.

NOTE 2 TDQS_t is not valid for the x4 configurationNOTE 3 TDQS_c is not available for the x4 configurationNOTE 4 DM_n not valid for the x4 configurationNOTE 5 For single die package (SDP) or 3DS, this pin is TEN; CS1_n is not connected, and is used as Test Mode Enable. For dual die package (DDP), this pin is CS1_n; and TEN is disconnected, and is used as CS1_n.

NOTE 6 The electrical parameters for DDP package are TBD and will be covered from future ballouts.

NOTE 7 For dual die package (DDP), ZQ is connected to bottom die (Rank 0).

NOTE 8 For single die package (SDP) or 3DS, CAI is used to indicate Command Address Inversion (ZQ1 is not applicable). However, for dual die package (DDP), CAI die pads are grounded inside the package, and ZQ1 pin is connected to the top die ZQ pad (Rank 1).

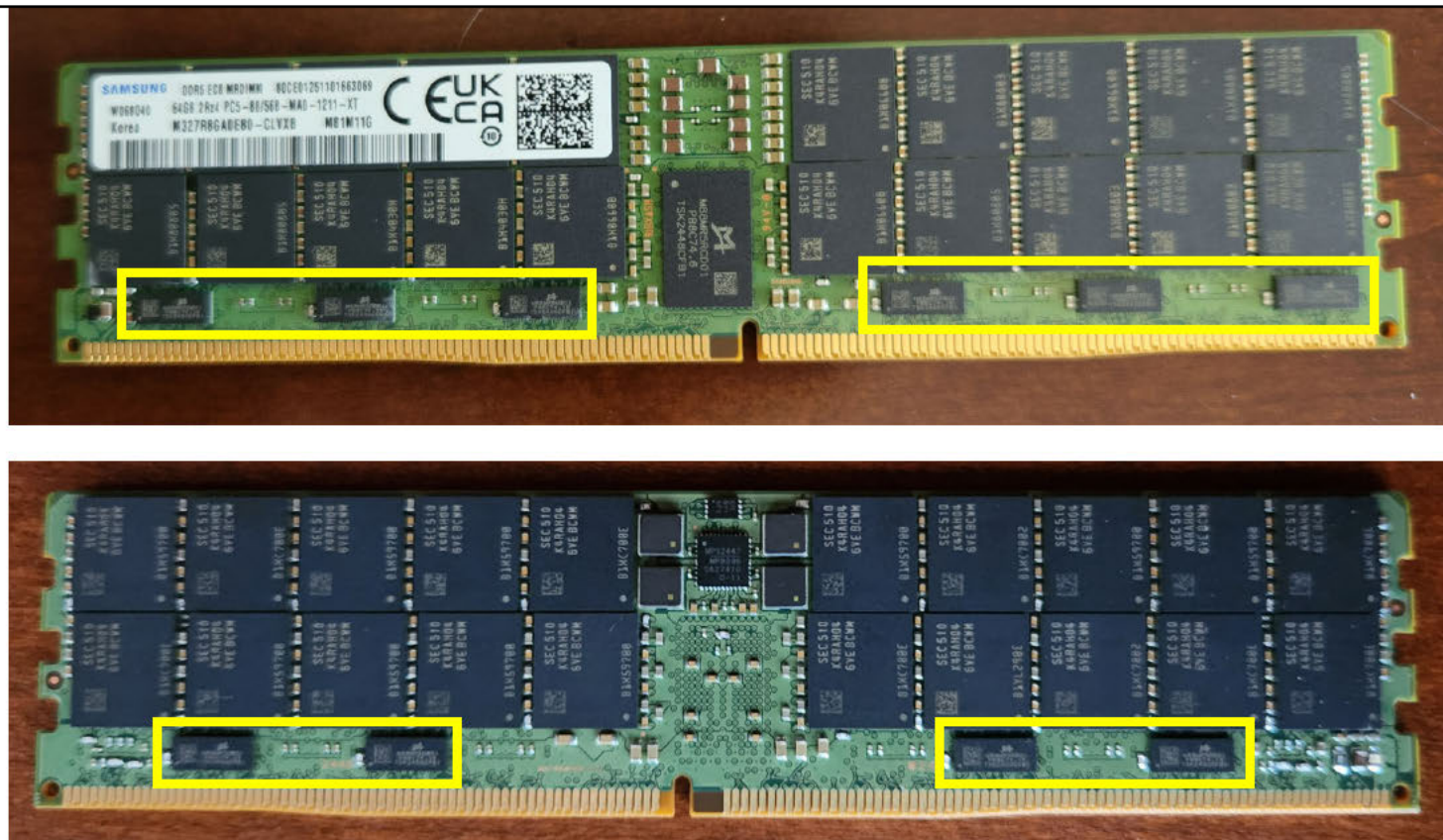
[1.c] [i] a data module mounted on the circuit board and [ii] coupled between the data ports of the memory

[i] Samsung's MRDIMMs each include a data module, such as a collection of ten MDBs (highlighted in yellow boxes), mounted on the circuit board, as illustrated below. As seen below, six of the MDBs are on the front side of the circuit board and four are on the back side.

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devices and the system memory bus,



Ex. CC-111 (Samsung MRDIMM M327R8GA0EB0-CLVXB) (annotated); Ex. CC-112 (20250811_151215.jpg) (annotated).

See also Ex. CC-22 (Montage M88MR5DB01) (listing the product information for Montage's MDB).

[ii] The data module, i.e., the collection of MDBs, is coupled between the data ports of the memory devices and the system memory bus.

As already noted, Montage's MDB states the "MDB chip buffers and re-drives data signals from the memory controller or DRAM devices" and it "efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal." Ex. CC-

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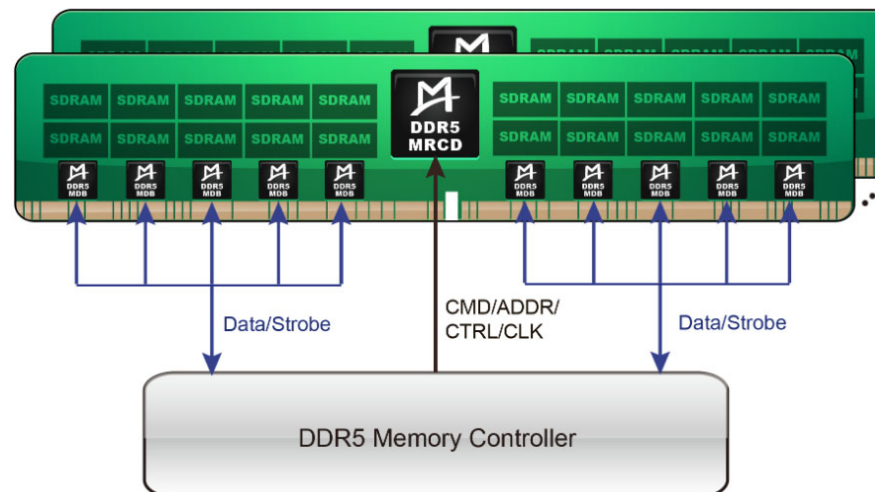
22 (Montage M88MR5DB01). Additionally, the MDB features “Dual nibbles on the host side,” “Four nibbles on the DRAM side, two per pseudo-channel,” and “Two DRAM-side DQ signals multiplexed to one host-side DQ signal.” *Id.*

This suggests that each data buffer is coupled between a respective set of memory devices and the memory controller, as illustrated on the Montage website:

Product Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.



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	Feature • Speed up to 8800 MT/s • Dual nibbles on the host side • Four nibbles on the DRAM side, two per pseudo-channel • Two DRAM-side DQ signals multiplexed to one host-side DQ signal • One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD) • x4 and x8 DRAMs support • Strobe and data training support • DRAM periodic update support • Multiple power-saving modes • Transparent, continuous burst, loopback and pass-through modes support • 1.1V VDD voltage • Green package: 78-Ball FBGA
[1.c] [iii] the data module including data handler logic elements; and	The data module on each of Samsung's MRDIMMs includes data handler logic elements. For example, the MDB states that it supports strobe and data training and different operating modes. This suggests that the data buffer has logic elements for data handling. Furthermore, Montage³ states that it is the "spec. owner of DDR5 . . . MDB . . . chips." Ex. CC-62 (Montage HK Proof) at 163. Thus, Montage's MDBs likely implement features in JEDEC's MDB01 Specification, such as [REDACTED] See, e.g., Ex. CC-18C (MDB01 Spec. Rev. 0.7) [REDACTED]

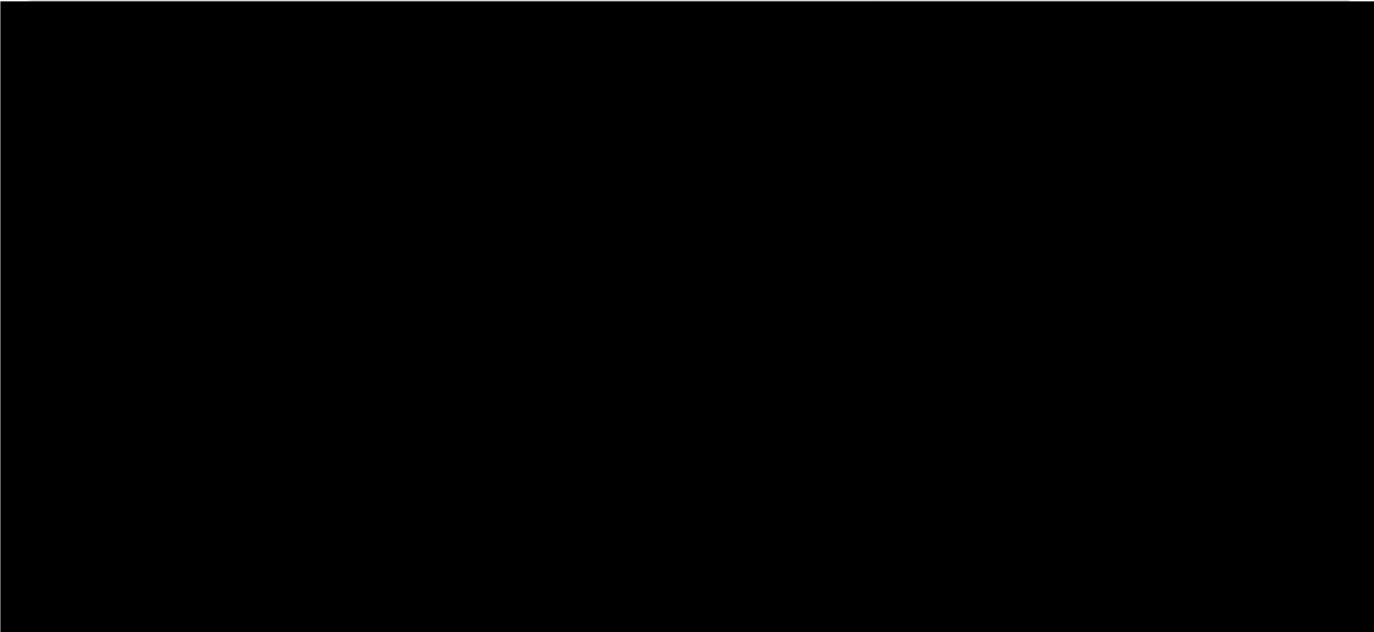
³ Montage is a JEDEC member. See Ex. CC-147 (JEDEC Member List).



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[1.d] [i] a control module mounted on the circuit board and [ii] coupled to the [a] data module, [b] the address and control ports of the memory devices, and [c] the system memory bus; and	[i] Samsung's DDR5 MRDIMM each contains a control module, such as the MRCD, mounted on the circuit board, as illustrated below:  Ex. CC-111 (Samsung MRDIMM M327R8GA0EB0-CLVXB).

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[ii.a] The control module (MRCD) is coupled to the data module. For example, one feature of Montage's MRCD is "Integrated BCOM bus to control the multiplexed data buffers (MDB)." Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer). The MRCD is therefore coupled to the MDBs in the data module via the BCOM bus to control their operation. *See also, e.g.,* Ex. CC-22 (Montage M88MR5DB01) (Montage's MDB features "One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)", which would be used to receive the control signals outputted by the MRCD).

[ii.b] The MRCD is also coupled to the address and control ports of the memory devices. For example, as noted above, the MRCD01 chip from Montage is designed to "buffer[] and re-drive[] the address, command, clock and control signals from the memory controller." Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer). These signals are understood to be driven to at least the memory devices in order to perform "efficiency memory management operations."

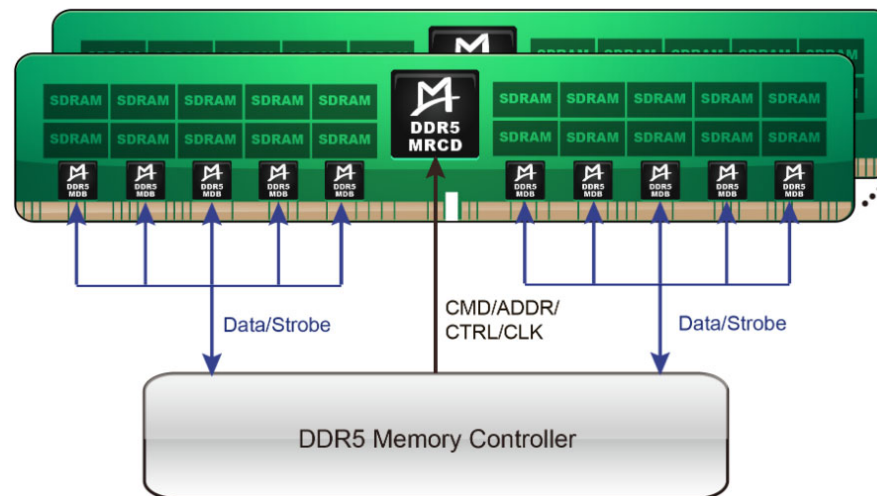
General Description

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General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in high-performance computing scenarios, such as AI training, inference, and large-scale data analysis.



(showing MRCD coupled to DDR5 Memory Controller).

[1.e] wherein the memory module is

Samsung's MRDIMMs are operable in any of a plurality of modes including a first mode and a second mode.

[REDACTED]

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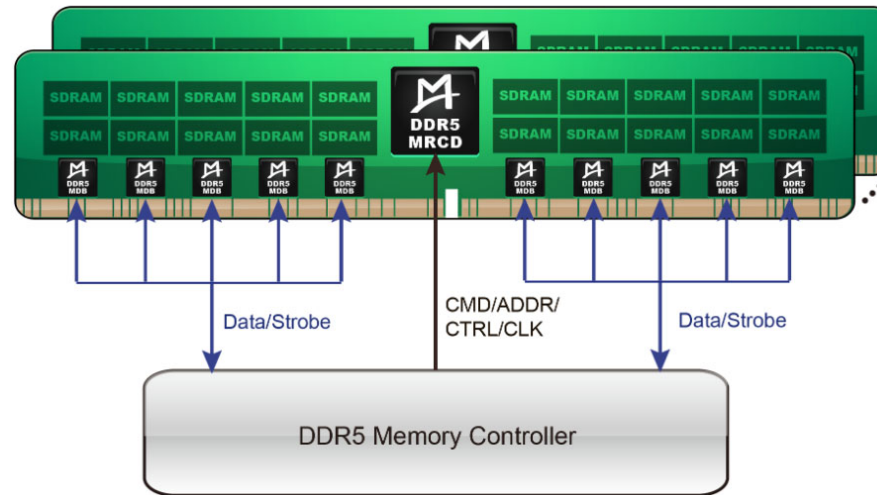
operable in any of a plurality of modes including a first mode and a second mode;	For example, the MDB supports and is therefore operable in a normal read/write operation mode. Ex. CC-22 (Montage M88MR5DB01) (“The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.”). The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal. As another example, in addition to the normal operating mode (an example of a first mode), Montage’s MDB also supports “[s]trobe and data training” (an example of a second mode), “[m]ultiple power-saving modes,” and “[t]ransparent, continuous burst, loop-back and pass-through modes.” Ex. CC-22 (Montage M88MR5DB01). Feature • Speed up to 8800 MT/s • Dual nibbles on the host side • Four nibbles on the DRAM side, two per pseudo-channel • Two DRAM-side DQ signals multiplexed to one host-side DQ signal • One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD) • x4 and x8 DRAMs support • Strobe and data training support • DRAM periodic update support • Multiple power-saving modes • Transparent, continuous burst, loopback and pass-through modes support • 1.1V VDD voltage • Green package: 78-Ball FBGA Additionally, under the JEDEC standards, it is expected that each of Samsung’s DDR5 MRDIMMs have an “[REDACTED]” (CC-18C, § 8.2; CC-24C, § 8.2), “[REDACTED]” (CC-18C, § 8.3; CC-24C, § 8.3), and “[REDACTED]” (CC-18C, § 3.10; CC-24C, § 3.14), which are
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	all related to training and testing. <i>E.g.</i> , CC-18C, § 3.10 (including a [REDACTED]); CC-24C, § 3.14 (same). Each of these modes satisfy the claimed requirements of the “second mode.”
[1.f] [i] wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals,	In each of Samsung’s DDR5 MRDIMMs, the control module, <i>i.e.</i>, the MRCD, in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals. For example, Montage describes its MRCD as handling “buffer[ing] and re-driv[ing] the address, command, clock, and control signals from the memory controller,” which means the MRCD is designed to receive the address/command/clock/control signals from the memory controller for otherwise it would not be able to buffer or re-drive those signals. <i>See</i> Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer). General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in high-performance computing scenarios, such as AI training, inference, and large-scale data analysis.

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(showing MRCD coupled to DDR5 Memory Controller).

As also explained, the control module (MRCD) is also configured, in the first mode (normal read/write), to output first memory address and control signals to the memory devices according to the system address and control signals. For example, Montage describes its MRCD as including the following features, where the incoming control/address inputs are split and separate control/address outputs are then generated based on the split incoming C/A signals:

- Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system
- Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs

Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer).

Additionally, Ex. CC-62 (Montage HK Proof) defines MRCD as:

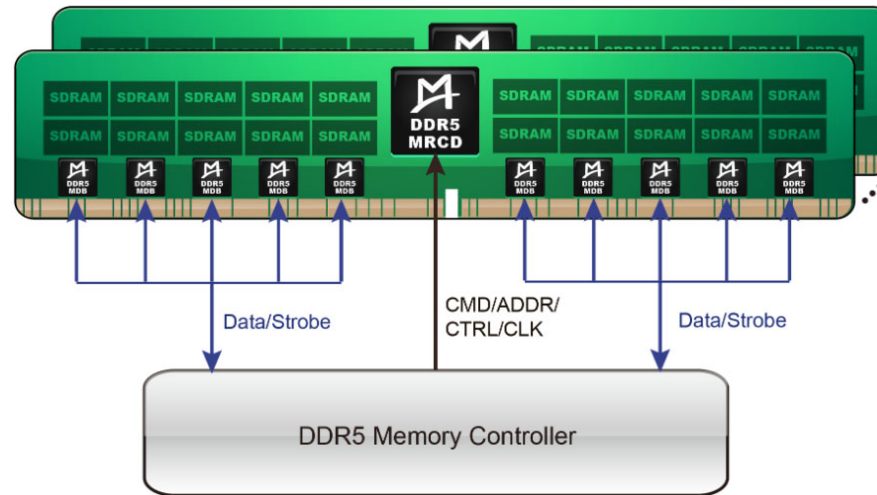
[REDACTED]

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	“MRCD” multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities <i>Id.</i> at 43. The signals re-driven across the multiple memory ranks of memory devices are those received and buffered from the system memory controller.
[1.f] [ii] and the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller,	In the first mode (normal read/write), the data module (the collection of ten MDBs) is configured to propagate one or more first data signals between the memory devices and the system memory controller. As already noted, Montage’s MDB chips each “buffers and re-drives data signals from the memory controller or DRAM devices” and “efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01). This suggests that the data buffers are configured to propagate one or more data signals between the memory devices and the system memory controller: Product Description The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal. Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.

[REDACTED]

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Feature

- Speed up to 8800 MT/s
- Dual nibbles on the host side
- Four nibbles on the DRAM side, two per pseudo-channel
- Two DRAM-side DQ signals multiplexed to one host-side DQ signal
- One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)
- x4 and x8 DRAMs support
- Strobe and data training support
- DRAM periodic update support
- Multiple power-saving modes
- Transparent, continuous burst, loopback and pass-through modes support
- 1.1V VDD voltage
- Green package: 78-Ball FBGA

[1.f] [iii] the one or more first data signals being transmitted or

In each of Samsung's MRDIMMs, the one or more first data signals are being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals. For example, in DDR5, commands for memory operations are defined by a combination of C/A and CS signals. *See* Ex. CC-46 (DDR5 Command Truth Table) (showing that commands for Activate, Read, and Write are composed of C/A[0:13] signals):

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received by at least a portion of the memory devices in response to the first memory address and control signals; and

Function	Abbreviation	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,17,20,21, 23,24
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,15,18,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,15,18,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DOPID	8,11,13,20, 22,23,24
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DOPID	8,13,20,22, 23,24,25,26
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,12,15,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,12,15,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,15,19,20, 21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,15,19,20, 21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3	

See also, e.g., Ex. CC-69 ((LP)DDR5 Memory Interface Presentation) at 11 (“[a]ll SDRAM commands are defined by states of command bus”).

The read and write operation and the associated transmission and receipt of one or more data signals are in response to the first memory control and address signals received by the memory devices. For example, whether the memory devices read or write operations depend on the combination of signals transmitted on the control/address and CS pins

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(examples of “the first memory address and control signals”). Depending on the value of chip select signal, a portion of or all of the memory devices may participate in the data transmission and receipt.

Cf., Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 11:

Symbol	Type	I/O Levels	Function
CS[1:0]_A CS[1:0]_B	Input	VDD	Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.

Ex. CC-13 (JEDEC DDR5 MRDIMM Plan) (a planned feature of MRDIMM is “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).

Ex. CC-58 (Kim 2020) at 168 (As general DDR5 technical background, a SK Hynix article illustrating basic functional blocks of a DDR5 SDRAM, showing that read/write control is in response to CS and CA signal inputs to the SDRAM device which signals collectively define commands for memory operations):

Fig. 2. DDR5 functional block diagram.

Each of Samsung's MRDIMMs is expected to have the same, or substantially the same, building blocks as reflected in this diagram.

In each of Samsung's MRDIMMs, the control module (MRCD) in the second mode (*e.g.*, training mode) is configured to output a second memory address and control signals to the address and control ports of the memory devices. For example, Montage, being the "spec. owner of DDR5 . . . MDB . . . chips" (Ex. CC-62 (Montage HK Proof) at 163),

[REDACTED]
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mode is configured to output second memory address and control signals to the address and control ports of the memory devices,	likely implements features in MDB01 and MDB02 Specifications. Montage supports data and strobe training. Ex. CC-22 (Montage M88MR5DB01). [REDACTED]. See Ex. CC-18C (MDB01 Spec. Rev. 0.7) [REDACTED] [REDACTED] [REDACTED]
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	[REDACTED] [REDACTED] [REDACTED]
[1.g] [ii] and the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller	In Samsung's MRDIMMs, in the second mode, the data module comprised of the collection of the ten MDBs is configured to isolate the memory devices from being accessed by the system memory controller. For example, [REDACTED] See Ex. CC-18C (MDB01 Spec. Rev. 0.7) [REDACTED] [REDACTED]

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	[REDACTED] CC-18C, [REDACTED] CC-24C, [REDACTED] [REDACTED] [REDACTED] [REDACTED] CC-18C, [REDACTED] CC-24C, [REDACTED]
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[REDACTED]
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	[REDACTED] [REDACTED] [REDACTED] CC-18C, [REDACTED] CC-24C, [REDACTED]
[1.g] [iii] and to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module,	In the training mode, in each of Samsung's MRDIMMs, the data module (collection of MDBs) is configured to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module. As noted above, [REDACTED]

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	[REDACTED] Ex. CC-18C (MDB01 Spec. Rev. 0.7) [REDACTED] [REDACTED] <i>Id.</i> [REDACTED] [REDACTED]
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	The host's write commands are first received by the MRCD and then output from the MRCD and sent to the MDB via the BCOM bus. <i>See</i> Ex. CC-22 (Montage M88MR5DB01) (the MDB has "[o]ne input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)"); Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) (MRCD has an "[i]ntegrated BCOM bus to control the multiplexed data buffers (MDB)"); <i>see also</i> Ex. CC-18C (MDB01 Spec. Rev. 0.7) CC-18C, CC-24C,
[1.g] [iv] and wherein at least a portion of the	In the training mode, in each of Samsung's MRDIMMs, at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory address and control signals from the control module.

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memory devices are configured to receive the one or more second data signals according to the second memory address and control signals from the control module.

As noted above, during MWD training,

[REDACTED]

[REDACTED]

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Ex. CC-18C (MDB01 Spec. Rev. 0.7)

Id.

CC-18C

CC-24C,

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	<i>See also</i> Ex. CC-62 (Montage HK Proof) at 43 (“MRCD” is used to “buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks”): <table><tr><td>“MRCD”</td><td>multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities</td></tr></table>	“MRCD”	multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities
“MRCD”	multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities		